EMS619U-BIOMEDICAL DEVICE DEVELOPMENT

COURSEWORK 2 – SPINAL IMPLANT DESIGN

GROUP B

Medical Device Development for Metastatic Spinal Cord Compression in Late-Stage Breast Cancer

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EXECUTIVE SUMMARY

Our case is a 40-year-old woman with late-stage breast cancer which has metastasised to her spine. This diagnosis is accompanied with reduced sensation, pain and impaired mobility. To ensure our patient receives the best treatment while also reducing hospital stay, our team have developed a novel balloon kyphoplasty device. This device tackles issues such as thrombosis and cement leakage to improve patient quality of life and to reduce the risk of additional symptoms. The device features a biodegradable balloon made from PLGA copolymer or PLA, aerosolised spray application of bone cement, hydrogel coating of balloon and the use of a stainless-steel stent. To ensure our device restores vertebral height and maintains stability under loading, a finite element method (FEM) analysis was carried out, which presented a deformation of 0.0033mm. By combining a biodegradable balloon with a controlled cement delivery, this device offers an effective and safer alternative for managing MSCC.

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1. INTRODUCTION AND BACKGROUND

The human spine is responsible for supporting body weight, protecting the spinal cord, and allowing flexible movement. It consists of cervical, thoracic, lumbar, sacral, and coccygeal regions (Figure 1), each with distinct roles in posture, mobility, and load transmission (Mahadevan, 2018). The spine is subjected to several complex loading conditions (Figure 2) including axial compression, bending, torsional, and shear forces during daily activities such as walking, sitting, and lifting. According to research, compressive loads in the lumbar region can exceed 2.5 times body weight during regular activities, reach 8 times body weight during lifting or bending, and increase dramatically during flexion, extension, or unsupported sitting (Wilke et al., 1999; Rohlmann et al., 2006).

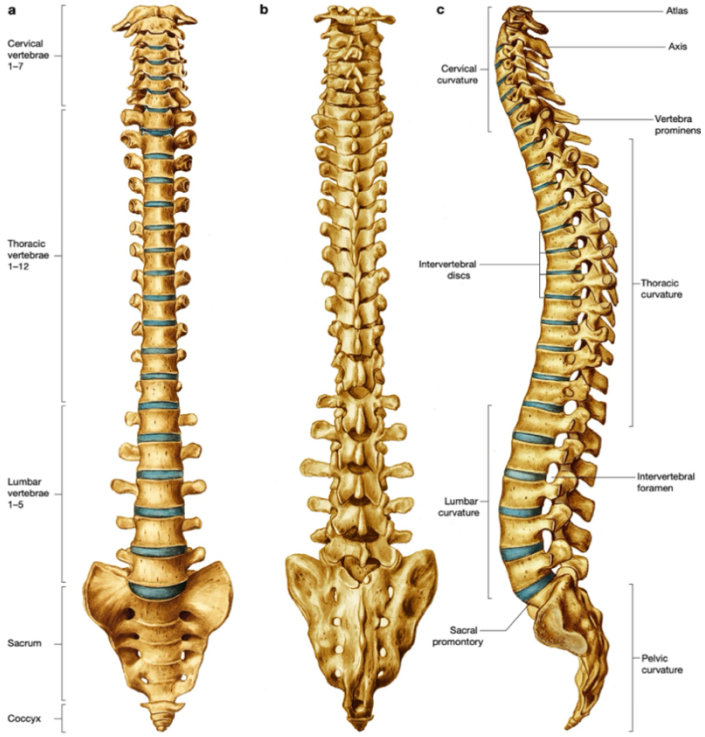


Figure 1: *Anterior* ***(a),*** *posterior* ***(b)*** *and lateral* ***(c)*** *views of an. adult articulated vertebral column* ***(Mahadevan, 2018).***

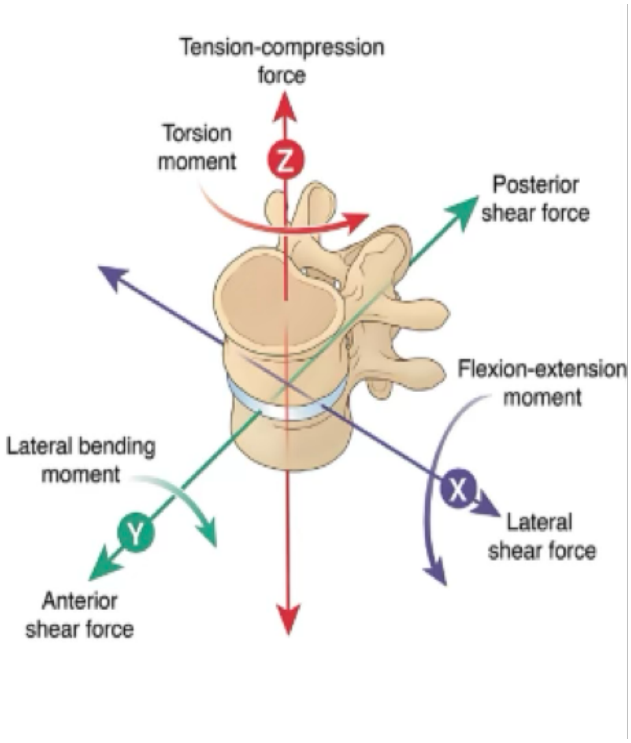


Figure 2*: Biomechanical coordinate system and direction of forces and moments* ***(Ko, 2023).***

Metastatic Spinal Cord Compression (MSCC) is an oncological emergency that affects up to 5–10% of patients with advanced cancer (Al-Qurainy and Collis, 2016; Lawton et al., 2018). It typically occurs when cancer metastasises to vertebral bodies (the anterior part of each vertebra responsible for weight-bearing), causing destruction and collapse of the vertebral body, which then compresses the dural sac, nerve roots, and the spinal cord itself. This compression leads to decrease in blood flow to the spinal cord and demyelination, causing progressive pain, motor weakness, sensory loss, and sphincter dysfunction over time. These symptoms often follow a gradual course, which can delay diagnosis (Boussios et al., 2018). The lumbar spine is especially vulnerable to metastasis due to its rich vascular network, and when vertebral integrity is compromised in cases such as MSCC, vertebral discs lose their ability to absorb and distribute loads, resulting in increased stress on the vertebral bodies. Spinal cord compression can result in irreversible functional loss in such cases (Ribas and Schiff, 2012). Any stabilising implant must thus account for these forces while ensuring long-term biomechanical compatibility with weakened tissue (Rohlmann et al., 2006; Galbusera et al., 2011). Vertebral body collapse in MSCC also commonly results in spinal misalignment and decrease in vertebral height. Stabilising implants must thus also aim to restore vertebral height to relieve spinal cord compression (Boussios et al., 2018).



Figure 3: *MRI of patient's lumbar spine showing L1*

This project addresses a clinical case involving a 40-year-old woman with late-stage breast cancer, presenting with MSCC due to damage caused by metastases at the L1 vertebra (Figure 3). The patient suffers from persistent pain, impaired mobility, and reduced sensation. This report presents the development and justification of a novel spinal

implant designed to stabilise the L1 vertebra in this context, and outlines the clinical rationale, engineering approach, and performance specifications of the design, as well as an analysis of the design's feasibility for prototype manufacture. The scope of this project includes a full design cycle, from conceptual ideation to the detailed specification of a device suitable for prototype manufacture.

2. DESIGN IDEAS AND CONCEPTS

2.1 Initial Concepts

A range of spinal implants and intervention strategies were explored to address the clinical challenge. These design options aimed to stabilise the patient's spine, relieve spinal cord compression, and align with the patient's palliative care needs. The solutions considered included:

- Laminectomy and spinal fusion (gold standard)
- Balloon kyphoplasty and vertebroplasty
- Pedicle screws with metal rods for stabilisation

2.2 Evaluation Criteria

Each design concept was assessed using a set of criteria relevant to the patient's clinical needs. Criteria were ranked by importance, with justifications summarised in Table 1 below:

***Table 1**: Ranking and Justification of Requirements*

Evaluation Criteria	Description	Importance (1–5)
Spinal Decompression & Stability	Implant must reinforce L1 vertebra to prevent further collapse and spinal cord compression.	5
Biocompatibility	Materials must avoid immune/inflammatory response to reduce implant rejection and revision surgery.	5
Tumour Control & Pain Management	Critical for improving patient's remaining quality of life.	4
Compatibility with Cancer Treatments	Implant must not interfere with radiotherapy or chemotherapy.	4
Surgical Invasiveness	Must be minimally invasive to reduce tissue damage, pain, and recovery time.	4
Medical Imaging Compatibility	Important for follow-up scans and radiotherapy planning.	3
Durability	Required to maintain structural integrity throughout patient's remaining life span.	3
Post-Implant Care	Simplified post-operative care improves patient comfort and reduces complications.	3

2.3 Concept Scoring

A weighted decision matrix (Table 2, in the appendix) was developed to compare the proposed design solutions. Each concept was evaluated based on its performance against the criteria using a scoring scale (1 = bad, 2 = good, 3 = better). The unweighted scores were multiplied by the assigned importance weightings to determine a total weighted score for each design.

2.4 Selected Concept

The selected design solution was **Balloon Kyphoplasty and Vertebroplasty**, which achieved a total weighted score of **66**, ranking above standard fusion techniques and metal rod stabilisation.

We chose BKP because it met the highest-priority requirements—namely spinal decompression & stability, minimal surgical invasiveness, and adequate compatibility with ongoing cancer treatments—better than the other concepts. Its minimally invasive nature was seen as advantageous in palliative care, whilst the short recovery time and potential to quickly relieve pain made it suitable for the 40-year-old patient with metastatic breast cancer. Although BKP does not inherently address the underlying tumour, it does significantly reduce fracture-related pain and stabilizes the vertebra. Moreover, as discussed in Section 3.1, BKP can be combined with tumour-targeting methods (e.g., RFA) or advanced cements to enhance local tumour control if needed.

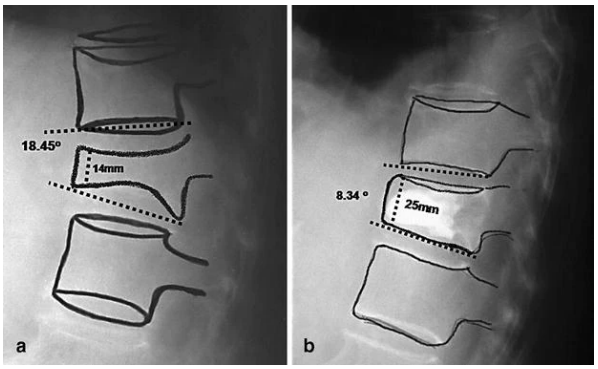


Figure 4: Radiographs showing the correction of vertebra collapse pre and post op ***(I. Gaitanis et al., 2005)***

Balloon Kyphoplasty is a minimally invasive procedure used to restore vertebra height. This procedure involves the use of bone cement, the most common one being PMMA. Initially a small incision is made using a fluoroscopic guide, enabling cannula insertion. The balloon catheter is inserted via the cannula and the balloon is inflated to a pressure above 200-250 psi. The insertion of the balloon creates a cavity within the vertebra and once removed, bone cement is slowly injected into the cavity (Patel, Petrone and Carter, 2023).

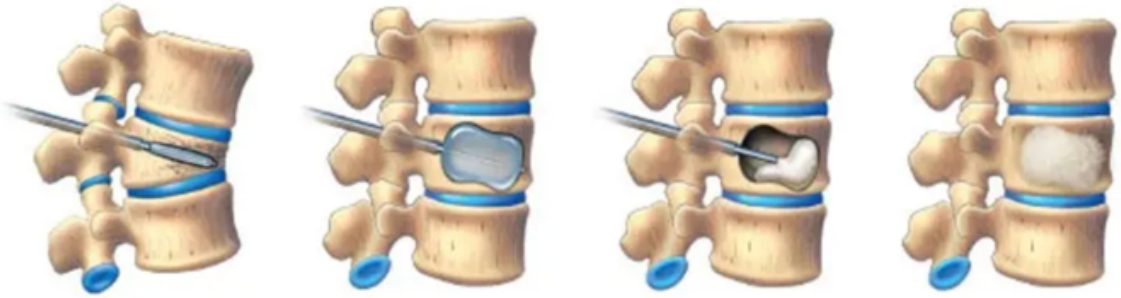


Figure 5: Balloon Kyphoplasty Procedure ***(Admin, 2019)***

The clinical studies for balloon kyphoplasty offer several advantages for patience with MSCC. This procedure offers spinal stabilisation and pain relief while also providing a short recovery time. However, despite these advantages, this treatment does not treat the underlying tumours causing the vertebra compression. Table *3 and 4* summarises the benefits and drawbacks of balloon kyphoplasty regarding our patient. Table *5* outlines the balloon related risk correlated with balloon kyphoplasty. A review of 75 patients observed complications such as cortical/endplate fractures due to balloon expansion, vertebral re collapse after balloon deflation, balloon rupture during inflation and transient hyperalgesia which occurs due to the exothermic properties of the bone cement (G Saliou et al., 2009). These complications highlight the importance of bone cement selection, precise balloon placement and controlled inflation.

2.5 Current innovations on balloon kyphoplasty

Recent refinements to Balloon Kyphoplasty have focused on reducing cement leakage, improving tumour control, and mitigating neurological compression (He et al., 2015; Li et al., 2020). Cement extravasation is minimized via higher-viscosity polymethylmethacrylate (PMMA) injected at a doughy phase, negative-pressure or “decompressed” injection techniques, and implant-based systems that channel the cement more precisely within the vertebral body (Pflugmacher et al., 2008; Hulme et al., 2006).

To address the underlying tumour burden, Balloon Kyphoplasty can be paired with minimally invasive tumour ablation approaches. These being radiofrequency ablation, experimental cements embedded with cytotoxic agents or even magnetic nanoparticles which are currently under investigation for localized therapeutic effects (G Saliou et al., 2009; Li and Mooney, 2016). In cases where epidural tumour extension threatens neural elements, a hybrid strategy involving decompression prior to BKP can reduce spinal canal pressure whilst maintaining vertebral stability and neurological function (Boussios et al., 2018).

2.6 Benefits, Drawbacks and Complications of Balloon Kyphoplasty

Table 3: Benefits of balloon kyphoplasty

Benefit	Description	Reference
Minimally Invasive	Reduces blood loss, healing time, infection risk, and hospitalisation duration.	Pflugmacher et al., 2008
Spinal Stabilisation	Supports the L1 vertebra, reducing the risk of further collapse.	Pflugmacher et al., 2008
Pain Relief	Removes compression of the spinal cord, limiting the need for opioid analgesics.	Pflugmacher et al., 2008
Functional Improvement	The Oswestry Disability Index (ODI) improved from 81 to 39 postoperatively.	Pflugmacher et al., 2008
Hospital Duration	Patients are typically discharged within 2 days after the procedure.	Pflugmacher et al., 2008

Table 4: Drawbacks of balloon kyphoplasty

Drawback	Description	Reference
Cement Leakage	Cement leakage can worsen neurological symptoms.	Pflugmacher et al., 2008
Not Cancer Targeting	Balloon kyphoplasty stabilises the vertebra but does not reduce the tumour or any tumour-related pain.	Pflugmacher et al., 2008
Adjacent Fracture Risk	Not common; occurred in 5 patients within 8 months.	Pflugmacher et al., 2008

Table 5: Balloon related complications

Complication	No. of Cases	Description	Reference
Cortical or Endplate Fracture	4 vertebrae	Occurs due to balloon overexpansion in vertebral endplates, especially within thin intervertebral disks.	G. Saliou et al., 2009
Vertebral Re-collapse	4 vertebrae	Occurs after balloon deflation, specifically if the patient has osteonecrosis.	G. Saliou et al., 2009
Balloon Rupture During Inflation	5 vertebrae	May cause contrast leakage. Risk reduced by slow inflation and purging air from the balloon. Usually not harmful.	G. Saliou et al., 2009
Transient Hyperalgesia	11 patients	Reoccurrence of pain during early stages of recovery; occurred in < 3%. Treated with steroids and nonsteroidal anti-inflammatory drugs.	G. Saliou et al., 2009

2.7 Bone Cement Usage

PMMA is the most common type of bone cement used in orthopaedics. As shown in table 6, it has desirable properties such as biocompatibility, and its high compressive strength. However, to improve the overall properties, additives have been added to PMMA such as $BaSO_4$ to make it radiopaque and Gentamicin/Vancomycin to introduce antibiotic properties (Arora, 2013). However, PMMA-Fe_3O_4 remains the preferred option for patients with MSCC due to its thermal reduction. The exothermic reaction during polymerisation generates heat, allowing for local thermal ablation. This helps target the surrounding tumours near the collapsed L1 vertebrae, allowing for stabilisation and localised tumour control (Yu et al., 2019).

Table 6: Comparison of different bone cements

Bone Cement Type	Base Material	Advantages	Disadvantages	Reference
PMMA	Polymethylmethacrylate	High compressive strength, biocompatible	Exothermic reactions during polymerisation	He et al., 2015
Antibiotic-loaded PMMA	PMMA + Gentamicin/Vancomycin	Antibacterial properties reducing risk of infection	Weakens mechanical properties	He et al., 2015
Radiopaque PMMA	PMMA + $BaSO_4$ / ZrO_2	Enhanced radiopacity	Lowers mechanical strength	He et al., 2015
Calcium Phosphate Cements	Hydroxyapatite	Osteoconductive. Suitable for low-load applications	Low mechanical strength	He et al., 2015

3. FINAL DESIGN AND FUNCTIONALITY

3.1 Design Overview

Our proposed design is a modification to the conventional balloon kyphoplasty that addresses the risks of thrombosis due to bone cement leakage as well as loss in restored vertebral height due to balloon tamp inflation (fig 6). Instead of releasing bone cement into an open cavity within the vertebrae, the cement would be applied via an aerosolised spray that is used to inflate the balloon while simultaneously depositing layers of bone cement upon the inner surface of the

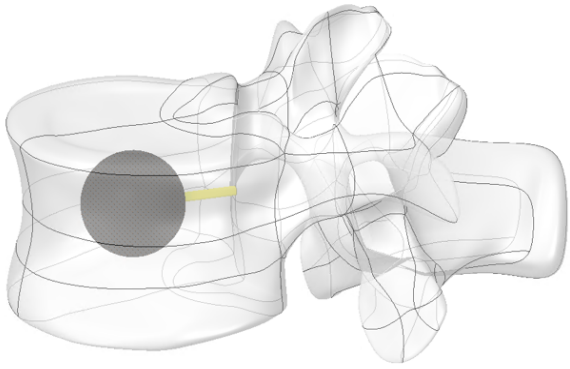


Figure 6: Model of our device inflated in the L1 vertebra.

balloon tamp until fully filled. To support the gradual deposition of bone cement within the balloon tamp during more sever unstable compression, an optional stent that is expanded along with the balloon provides rigid support.

The introduction of a biodegradable balloon made of PLGA copolymer or PLA that degrades after vertebral height restoration hence eliminating removal surgery (Liu and Ma, 2004). A hydrogel coating on the balloon surface would also enhance biocompatibility and deliver therapeutics for pain relief as discussed in a review article for hydrogels it was found that Hydrogels can provide spatial and temporal control over the release of various therapeutic agents, including small-molecule drugs, macromolecular drugs and cells (Hoffman, 2012). As PLA has a working temperature above the 56 degrees Celsius temperature of PMMA during exothermic curing reaction, it can hold the cement without severe degradation of its structural integrity (Stoops et al,2022).

Simultaneous cement injection during balloon inflation would aid in optimising cement distribution as well as reduce leakage risks as it was found in clinical trial to compare two procedural interventions cement directed kyphoplasty system (CDKS) and vertebroplasty and it was concluded from the trial that CDKS effectively reduces posterior cement leakage, reducing the risk of leakage related complications (Vogl et al., 2013). A critical innovation in the design also addresses the risk of bone cement aerosolization (particulate dispersion during injection). A closed-system cement delivery where the balloon's cement microchannels incorporate a diffuser tip to reduce high-velocity jetting of cement, Nanoporous filters at the nozzle exit break cement into larger droplets hence minimising fine particulate generation (Del et al., 2022). According to the research (Kumar et al., 2024) it can be determined that Ventilated

Ballon Design where the outer layer of the balloon includes a gas-permeable membrane to vent trapped air during injection – prevents pressure buildup that exacerbates aerosolization. The exothermic reaction of the PMMA cement curing is treated as an advantage as it can passively damage the remaining cancer tissue within the vertebrae.

A Stainless steel 316L permanent stent can be placed over the balloon tamp for decompression cases where damage to the vertebrae is severe enough that the restored vertebral heigh via decompression is unstable. Stainless steel stents provide the mechanical properties necessary to support the vertebrae post decompression, with stress shielding not considered a severe problem due to the expected lifetime of the device in the patient. With these modifications to the conventional technique, we aim at targeting the restoration of the L1 vertebral height, mitigation of the spinal cord compression as well as long term stabilisation.

4. PROCEDURE, ADVANTAGES, DISADVANTAGES, AND LIMITATIONS

4.1 General surgical approach

Balloon kyphoplasty is performed under local or general anaesthesia with fluoroscopic or CT guidance, with the patient positioned prone. A small incision is made in the patients to access the effected vertebra, in this case the L1 (Fig 3). A hand drill or curette is used to create a channel into the vertebral body. Through this channel, our device is placed on a catheter and is inserted gradually using a cannula. Once in position within the vertebrae, controlled expansion of the balloon tamp is performed with the aerosolised PMMA bone cement at a controlled pressure. Once the balloon tamp is fully expanded and vertebral height is restored the balloon tamp is filled with bone cement and left to cure and harden. The surgical instruments are then removed, and the incision is closed.



Figure 7 Inflated kyphoplasty Balloons (tamps) ***(Lindare medical (2019))***



Figure 8 Inflated Kyphoplasty Balloons (Tamps) ***(Lindare Medical (2019))***

4.2 Clinical Advantages

The no leakage ballon kyphoplasty device is designed to eliminate one of the most common complications of traditional kyphoplasty, which is cement leakage. Leakage has been reported in up to 13.7% of cases and can lead to severe outcomes like spinal canal invasion or even vertebral necrosis (Zhang et al., 2015). By preventing leakage, this device can reduce the need for reoperation and improve overall patient safety. Techniques that prioritise cement containment, such as the eggshell method, have already demonstrated that targeted delivery can get rid of symptomatic leakage entirely while improving patient outcome. For example, Lo (2017) reported no observed cement and improved vertebral alignment. In addition, improving procedural safety, controlling cement delivery may also enhance long-term comfort and mobility. Patients treated with ballon kyphoplasty were more likely to return full ambulation within one week and maintain social independence one-year post-treatment, compared to those managed conservatively (Klezle et al., 2012)

4.3 Manufacturing feasibility and limitations

While clinically promising, this device may face limitations in manufacturing. This added design complexity could increase production costs and require more advanced fabrication techniques. These could possibly affect accessibility, especially in lower-income healthcare systems. The cost of balloon kyphoplasty remains a significant barrier for it to be bought. Additionally traditional devices already rely on synthetic materials like PMMA, which pose environmental concerns due to their non-biodegradable nature and energy intensive production (Zhang et al., 2015).

4.4 Patient comfort and quality of life

Leakage control can also help patient outcomes, including reduced postoperative pain and disability. (Lo (2017)) found that techniques focused on cement containment led to significant improvements in pain scores and spinal alignment. This can suggest that a leakage-free system could further enhance mobility and independence in daily life. By minimising the risk of complications such as cement leakage into neighbouring tissues or the spinal canal, patients are less likely to achieve postoperative discomfort, nerve irritation, or the need for additional interventions. Also, the vertebral height restoration contributes not only to structural stability but also better posture and reduced muscular strain, which are key factors in long term comfort. When combined with a streamlined surgical process, this design can also reduce hospital stay duration, which can be very important for this patient as this patient may not have long to live hence allowing the patient normally at a quicker time. Because of how minimally invasive Balloon Kyphoplasty is, it has been associated with a significantly shorter hospital stay. Svedbom et al. (2012) estimated that Balloon kyphoplasty reduces inpatient time by an average of 6 days compared to non-surgical management, contributing to both faster patient recovery and reduces healthcare costs. This highlights the benefits of this procedure, enhanced patient comfort and efficiency in resource utilisation

5. INITIAL VALIDATION AND ANALYSIS

To validate the efficacy of our design in stabilising and supporting the compressive forces applied to the L1 vertebrae with minimal deformation and loss of vertebral height, an FEM simulation was performed using Ansys Static Structural. To simplify the model to account for the limited computational power available, the device is treated to be spherical when inflated, with solid layers enclosing the inner bone cement core (fig 9). The vertebra was treated to be composed of cancellous bone with Young's Modulus of 12,000MPa and a poison ratio of 0.3 (Dreischarf et al, 2014). The Ansys library was used to provide the material property data for the device components.

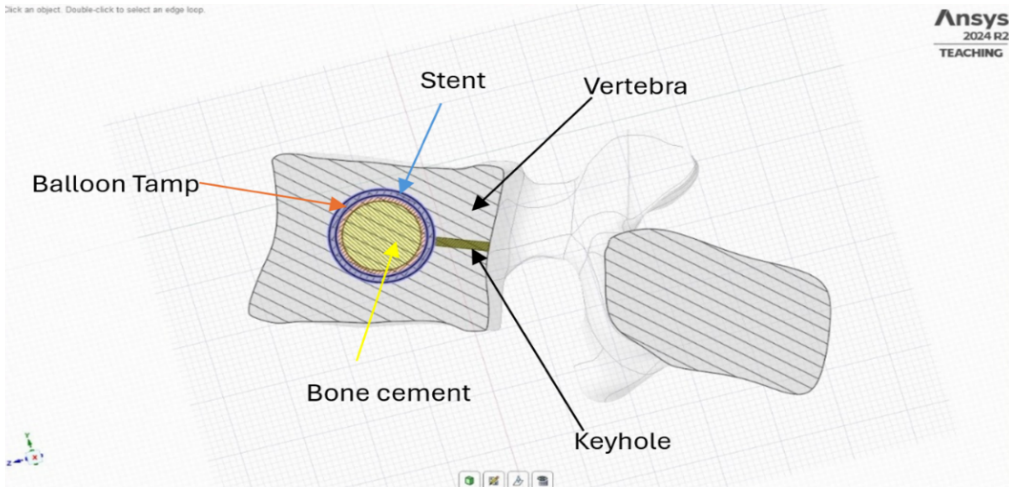


Figure 9: Cross section of the model used for FEM analysis

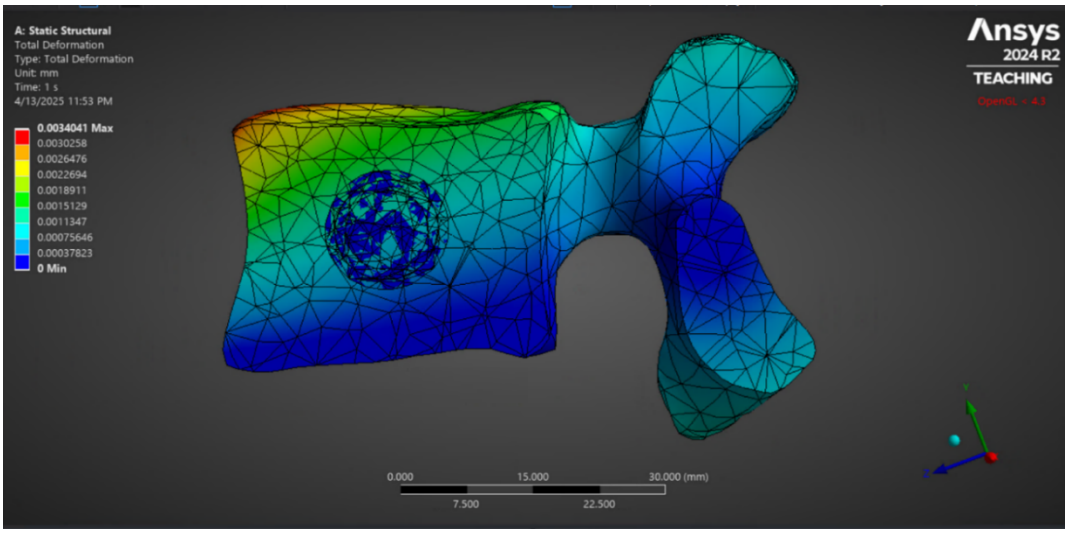


Figure 10: Cross section model overlaid with the FEM analysis results of the total deformation

Mesh sizing was set to 5e-4mm. For the boundary conditions the bottom surface of the L1 vertebrae is treated as a fixed support, with a compressive load of 1000N being applied to the top surface to represent typical lumbar compressive loads (Warren et al,2020). The FEM simulation solved for the total deformation of the model. The results show a maximum total deformation of 0.0034041mm away from the centre of the vertebral body which validates the stabilisation of our device in retaining the restored vertebral height (fig 10). This shows that with the device in place there is minimal loss in vertebral height post implantation – further supported by the device itself exhibiting little to no deformation under this load.

6. FUTURE DEVELOPMENT

Ongoing advances in biomaterials and surgical technology are driving the evolution of Balloon Kyphoplasty beyond simple fracture stabilization (Kim & Hwang, 2019; Zhang et al., 2021). Research increasingly focuses on bioactive and osteoconductive cements, targeted drug delivery systems, computer-assisted interventions to simultaneously improve structural support and therapeutic outcomes (Smith et al., 2020; Meyer et al., 2022), and minimally invasive decompression, which our biomedical device focuses on, allowing vertebral augmentation to be combined with oncologic and neurologic treatments (Yu et al., 2022). This integrated approach can improve local tumour control, enhance spinal cord decompression, and reduce cement-related complications (Meyer et al., 2022). Future strategies may also incorporate tumour ablation techniques (e.g., radiofrequency ablation) or cytotoxic/magnetic nanoparticle-embedded cements (Saliou et al., 2009; Li & Mooney, 2016), simultaneously reinforcing the vertebrae and delivering localized cancer therapy (Saliou et al., 2009; Li & Mooney, 2016). In cases where the tumour threatens neural structures, early decompression—potentially guided by robotic or computer-assisted systems—could preserve neurologic function and increase procedural precision (Yu et al., 2022).

6.1 Risks and Mitigations

The potential technical risks include balloon rupture or inadequate cement fill, and the clinical risks encompass infection, neurological complications, and cement leakage. These can be mitigated via controlled high-viscosity cement injection, real-time imaging guidance, and rigorous sterility protocols.

6.2 Regulatory and Ethical Considerations

The device first needs to be classified correctly to secure UK Conformity Assessed (UKCA) marking and register with the Medicines and Healthcare products Regulatory Agency (MHRA)—all of which involve adhering to a recognised quality management system and preparing robust risk analyses (Emergobyul, 2023a; Emergobyul, 2023b). This regulatory process must continue post-approval with ongoing surveillance, including reporting adverse events through the MHRA's Yellow Card scheme (NICE, 2023a). Equally crucial is fulfilling ethical obligations: clinical investigations in palliative care must gain approval from an independent Research Ethics Committee via the Health Research Authority, ensuring that the study design respects patient autonomy, obtains valid informed consent, and balances beneficence with non-maleficence (Gov.uk, 2023a; Gov.uk, 2023b; QICN, 2023). Finally, aligning the device's intended use with NICE guidelines for spinal metastases and metastatic spinal cord compression—by incorporating a multidisciplinary approach and clear governance structures—helps ensure the balloon kyphoplasty system is both safe and clinically appropriate (NICE, 2023b).

7. Conclusion

Balloon Kyphoplasty, traditionally used for vertebral fracture repair, was adapted to address both structural stabilization and local oncologic control for patients with metastatic spinal cord compression. By refining the technique—containing the bone cement within the expanded balloon tamp and optional stenting—we mitigated the key challenges of pain, neurological compromise, and tumour progression, particularly in our 40-year-old patient with advanced metastatic spinal cord compression. Moving forward, continued research on biomaterials, robotic or computer-assisted methods, and robust mechanical testing will be vital for ensuring broad clinical adoption. Ultimately, this design meets the central objectives of improving spinal stability, pain relief, and quality of life in late-stage cancer patients.

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APPENDICES

Table 2:: Design Decision Matrix

Criteria / Specification	Importance Weighting	Laminectomy and Spinal Fusion		Balloon Kyphoplasty and Vertebroplasty		Metal Rods and Screws Stabilisation	
		Unweighted	Weighted	Unweighted	Weighted	Unweighted	Weighted
Biocompatibility	5	2	10	3	15	3	15
Durability	3	2	6	1	3	3	9
Compatibility with cancer treatments	5	2	10	1	5	2	10
Surgical Invasiveness	4	2	8	3	12	1	4
Cost	1	2	2	3	3	3	3
Post implant care	3	2	6	3	9	2	6
Medical Imaging compatibility	2	2	4	3	6	1	2
Tumour Control & Pain Management	4	2	8	1	4	1	4
Spinal Decompression & Stability	5	2	10	2	10	2	10
Total Weighted Score			64		66		62